

Automated PVC Hot Water Coupling Using AI & Robotics

An Innovative Approach to Industrial Automation

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Current Process & Challenges

- PVC hot water coupling traditionally requires manual labor with several critical steps
- Process involves dipping pipes in hot water, precise alignment, and consistent pressure
- Manual method introduces variations in quality, workplace hazards, and production bottlenecks
- Limited scalability due to dependency on skilled human workers
- Safety concerns from exposure to hot water and chemicals

Problem Statement

- Current process is labor-intensive, introducing inconsistencies in product quality
- Existing automation solutions still require significant human intervention
- Semi-automated systems fail to address the need for a fully autonomous process
- Manual handling creates fatigue-related errors and safety hazards
- Quality control challenges due to variations in the manual process

Proposed Solution

- End-to-end automation system combining advanced robotics, computer vision, and adaptive force control
- Novel flat surface assumption for simplified automation and robotic movement
- 4-DOF robotic arm with 3-finger adaptive gripper for precise pipe handling
- YOLOv8-based vision system for pipe detection and localization
- Advanced hierarchical inverse kinematics for precision placement

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Key Research Influences

- Shuwei Qiu & Mehrdad Kermani (2021): Hierarchical IK approach for precise positioning
- Hongwei Zhang et al. (2024): Constant-force grasping for delicate object handling
- Joosep Reimand et al. (2023): YOLOv8 integration for industrial robotics applications
- Lin Chen et al. (2022): Minimum jerk trajectory planning for smooth robotic movements
- Zhao & Adelson: GelSight Svelte Hand 3-finger design inspirations

Gap Analysis & Our Contributions

- Current systems lack integration of vision, force control, and precision kinematics
- Existing solutions typically address individual components rather than end-to-end systems
- We integrate compliant beam mechanics with hierarchical inverse kinematics
- Novel application of Orthogonal Procrustes Problem for pipe-mold alignment
- First implementation of adaptive force gripping specifically for thermoplastic materials

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System Architecture Overview

- Flat surface layout with centrally mounted camera for simplified automation
- 4-DOF robotic arm with downward-facing end effector and 3-finger adaptive gripper
- Modular system with four key components: vision, gripping, placement, and control
- Complete workflow: detection, picking, heating, alignment, and insertion
- YOLOv8 integration for real-time pipe detection and localization

YOLOv8 Object Detection

- Transfer learning from pre-trained weights for efficient adaptation
- Fine-tuning with custom dataset of 2,500 annotated PVC pipe images
- 2D to 3D coordinate conversion using camera parameters:
- $x = \frac{(center_x - c_x) \cdot depth}{f_x}$, $y = \frac{(center_y - c_y) \cdot depth}{f_y}$
- Robust performance across different lighting conditions (92% accuracy)

Adaptive Force-Optimized Gripping

Minimizing Maximum Contact Force for Safe and Reliable Grip

- **Objective:** Minimize the **maximum contact force**

$$\min \max |f_i|$$

- **Technique:** Using penalty function and gradient descent.
- **Key Constraints:**
 - ✓ **Force Balance:** Cancel external disturbances
 - ✓ **Friction Cone:** Prevent slip
 - ✓ **Torque Limits:** Respect mechanical safety limits

Consistent and gentle grip ensures safety in handling delicate objects.

Minimum-Jerk Trajectory Planning

- 5th-order polynomial for smooth, natural movements:
- $x(t) = x_0 + (x_f - x_0) \left[10 \left(\frac{t}{T} \right)^3 - 15 \left(\frac{t}{T} \right)^4 + 6 \left(\frac{t}{T} \right)^5 \right]$
- Zero velocity and acceleration at trajectory endpoints
- Multi-waypoint path planning through critical process points
- Minimized jerk reduces mechanical stress and vibration

Hierarchical Inverse Kinematics

- Damped Least Squares solution for singularity robustness:
- $\Delta\theta = J^T (JJ^T + \lambda^2 I)^{-1} e$
- Nullspace projection for task prioritization: $P = I - J^\dagger J$
- Priority order: pipe positioning $\hat{}$ joint limit avoidance $\hat{}$ energy minimization
- Sub-millimeter precision in final placement position and orientation

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Performance Metrics

- Throughput increased by 285% (from 20 to 77 units/hour)
- Defect rate reduced by 85.9% (from 8.5% to 1.2%)
- Placement accuracy improved from $\pm 2.0\text{mm}$ to $\pm 0.3\text{mm}$ (85% improvement)
- Labor hours reduced by 94% (from 50 to 3 hours per 1000 units)
- Setup time for size changes decreased from 45 to 8 minutes (82% reduction)

Detection & Force Control Results

- YOLOv8 vision system achieves 98.7% accuracy under optimal lighting
- Robust performance maintained at 92.8% accuracy with variable lighting
- Adaptive gripper maintains force variation within $\pm 3\%$ across all pipe sizes
- Sub-millimeter positioning accuracy (0.27mm mean error)
- 99.4% successful insertions with no pipe damage across 500 test cycles

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RoboDK Simulation

- Full process simulation from pipe detection to final placement
- Realistic modeling of robotic arm with 4 degrees of freedom
- Complete workflow demonstration including hot water bath and molding
- Precision visualization of inverse kinematics and path planning
- Integration of camera system and custom gripper in simulation environment

Mathematical Simulations

- 2D coupling process simulation for kinematic validation
- Visualization of compliance-based force control in gripper simulation
- Minimum jerk trajectories for smooth pipe movement paths
- Hierarchical IK solver demonstration showing task priorities
- Comparison of different control strategies and their effects

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Key Achievements

- Successful development of fully automated end-to-end PVC coupling system - model and simulation
- Integration of AI vision, adaptive gripping, and precision control
- Dramatic improvements in throughput, quality, and worker safety
- Reduced production costs through labor savings and material waste reduction
- Mathematical foundations established for compliant gripping and hierarchical control

Future Work

- Multi-size adaptability for wider range of pipe diameters and configurations
- Online learning for continuous optimization of gripping force and placement
- IoT integration for remote monitoring and predictive maintenance
- Development of collaborative version for human-robot shared environments
- Extension to other thermoplastic materials with different properties

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- [2] Zhang, H., et al. (2024). *Optimizing Contact Force on an Apple Picking Robot End-Effector.*
- [3] Reimand, J., et al. (2023). *YOLOv8-Based Real-Time Object Detection for Industrial Robotics.*
- [4] Chen, L., et al. (2022). *Minimum Jerk Trajectory Planning for High-Precision Robotic Operations.*
- [5] Zhao, J. & Adelson, E. H. *GelSight Svelte Hand: A Three-finger, Two-DoF, Tactile Robot Hand.*

- **Code Implementation:**

`https:`

`//github.com/KeerthivasanSV/Automation-in-Pipe-industry`

- **Documentation & Simulations:**

`https://github.com/KeerthivasanSV/Robotics-MFC-Report`

- Full simulation videos and demonstration materials available
- Complete mathematical derivations and implementations provided

Thank You

MFC ROBOTICS Group 9 (AIE "A" Batch)
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